

Consistency Questionnaire Results

Q1. Missing physiological attributes

■ No inconsistencies found.

Q2. Multiple HR per ActorState

■ No inconsistencies found.

Q3. Multiple HRV per ActorState

■ No inconsistencies found.

Q4. Multiple RR per ActorState

■ No inconsistencies found.

Q5. Multiple SpO2 per ActorState

■ No inconsistencies found.

Q6. Multiple Drowsiness per ActorState

■ No inconsistencies found.

Q7. Missing high-level attributes

■ 7 inconsistencies found.

Running Q7. Missing high-level attributes...

■ Inconsistencies found (5 rows)

```
(rdflib.term.URIRef('http://test.org/in_cabin_domain#DriverState_c0a187de39c7466bbf49ba0102afd70f'),)
(rdflib.term.URIRef('http://test.org/in_cabin_domain#DriverState_62b16285b0174646b1b5cb64aeb5aec2'),)
(rdflib.term.URIRef('http://test.org/in_cabin_domain#DriverState_006e88f707294d2597670c37d0ed996a'),)
(rdflib.term.URIRef('http://test.org/in_cabin_domain#DriverState_f7d1d628477b434997d6c903beed5604'),)
(rdflib.term.URIRef('http://test.org/in_cabin_domain#DriverState_895298e1179f4fd0a4e458b390457cde'),)
```

Q8. Multiple Fatigue per ActorState

■ No inconsistencies found.

Q9. Multiple Attention per ActorState

■ No inconsistencies found.

Q10. Multiple Unresponsiveness per ActorState

■ No inconsistencies found.

Q11. Multiple EyeState per ActorState

■ No inconsistencies found.

Q12. Multiple MouthState per ActorState

■ No inconsistencies found.

Q13. Multiple Age or Sex per ActorState

■ No inconsistencies found.

Q14. Multiple Accessories/Demographic/FaceCharacteristics per ActorState

■ No inconsistencies found.

Q15-A. HR instances classified into more than one HR-level class

■ No inconsistencies found.

Q15-B. HRV instances classified into more than one HRV-level class

■ No inconsistencies found.

Q15-C. RR instances classified into more than one RR-level class

■ No inconsistencies found.

Q15-D. SpO2 instances classified into more than one SpO2-level class

■ No inconsistencies found.

Q15-E. Drowsiness instances classified into more than one Drowsiness-level class

■ No inconsistencies found.

Q16-A. HR without any HR-level classification

■ 7 inconsistencies found.

Running Q16-A. HR without any HR-level classification...

■ Inconsistencies found (5 rows)

```
(rdflib.term.URIRef('http://test.org/in_cabin_domain#hr_instance_2024_4_15_0_0_0'),)
(rdflib.term.URIRef('http://test.org/in_cabin_domain#hr_instance_2024_4_15_0_0_1'),)
(rdflib.term.URIRef('http://test.org/in_cabin_domain#hr_instance_2024_4_15_0_0_2'),)
(rdflib.term.URIRef('http://test.org/in_cabin_domain#hr_instance_2024_4_15_0_0_3'),)
(rdflib.term.URIRef('http://test.org/in_cabin_domain#hr_instance_2024_4_15_0_0_4'),)
```

Q16-B. HRV without any HR-level classification

■ 7 inconsistencies found.

Running Q16-B. HRV without any HR-level classification...

■ Inconsistencies found (5 rows)

```
(rdflib.term.URIRef('http://test.org/in_cabin_domain#hrv_instance_2024_4_15_0_0_0'),  
)  
(rdflib.term.URIRef('http://test.org/in_cabin_domain#hrv_instance_2024_4_15_0_0_1'),  
)  
(rdflib.term.URIRef('http://test.org/in_cabin_domain#hrv_instance_2024_4_15_0_0_2'),  
)  
(rdflib.term.URIRef('http://test.org/in_cabin_domain#hrv_instance_2024_4_15_0_0_3'),  
)  
(rdflib.term.URIRef('http://test.org/in_cabin_domain#hrv_instance_2024_4_15_0_0_4'),  
)
```

Q16-C. RR without any RR-level classification

■ 7 inconsistencies found.

Running Q16-C. RR without any RR-level classification...

■ Inconsistencies found (5 rows)

```
(rdflib.term.URIRef('http://test.org/in_cabin_domain#rr_instance_2024_4_15_0_0_0'),  
(rdflib.term.URIRef('http://test.org/in_cabin_domain#rr_instance_2024_4_15_0_0_1'),  
(rdflib.term.URIRef('http://test.org/in_cabin_domain#rr_instance_2024_4_15_0_0_2'),  
(rdflib.term.URIRef('http://test.org/in_cabin_domain#rr_instance_2024_4_15_0_0_3'),  
(rdflib.term.URIRef('http://test.org/in_cabin_domain#rr_instance_2024_4_15_0_0_4'),
```

Q16-D. SpO2 without any SpO2-level classification

■ 7 inconsistencies found.

Running Q16-D. SpO2 without any SpO2-level classification...

■ Inconsistencies found (5 rows)

```
(rdflib.term.URIRef('http://test.org/in_cabin_domain#spo2_instance_2024_4_15_0_0_0')  
,  
(rdflib.term.URIRef('http://test.org/in_cabin_domain#spo2_instance_2024_4_15_0_0_1')  
,  
(rdflib.term.URIRef('http://test.org/in_cabin_domain#spo2_instance_2024_4_15_0_0_2')  
,  
(rdflib.term.URIRef('http://test.org/in_cabin_domain#spo2_instance_2024_4_15_0_0_3')  
,  
(rdflib.term.URIRef('http://test.org/in_cabin_domain#spo2_instance_2024_4_15_0_0_4')
```

Q16-E. Drowsiness without any Drowsiness-level classification

■ 7 inconsistencies found.

Running Q16-E. Drowsiness without any Drowsiness-level classification...

■ Inconsistencies found (5 rows)

```
(rdflib.term.URIRef('http://test.org/in_cabin_domain#drowsiness_instance_2024_4_15_0_0_0'),  
(rdflib.term.URIRef('http://test.org/in_cabin_domain#drowsiness_instance_2024_4_15_0_0_1'),  
(rdflib.term.URIRef('http://test.org/in_cabin_domain#drowsiness_instance_2024_4_15_0_0_2'),  
(rdflib.term.URIRef('http://test.org/in_cabin_domain#drowsiness_instance_2024_4_15_0_0_3'),  
(rdflib.term.URIRef('http://test.org/in_cabin_domain#drowsiness_instance_2024_4_15_0_0_4'),
```

Q17. HRV instances with >1 HRV level

■ No inconsistencies found.

Q18. RR instances with >1 RR level

■ No inconsistencies found.

Q19. SpO2 instances with >1 SpO2 level

■ No inconsistencies found.

Q20. Drowsiness instances with >1 Drowsiness level

■ No inconsistencies found.

Q21. Violation "Sleep=> Inattentive & Unresponsive"

■ 3 inconsistencies found.

```
Running Q21. Violation "Sleep=> Inattentive & Unresponsive"...
■ Inconsistencies found (1 rows)
(rdfLib.term.URIRef('http://test.org/in_cabin_domain#DriverState_62b16285b0174646b1b5cb64aeb5aec2'), None,
 rdfLib.term.URIRef('http://test.org/in_cabin_domain#inattentive_instance'),
 rdfLib.term.URIRef('http://test.org/in_cabin_domain#undefined_atrisk_instance'))
```

Q22. Violation "Drowsy & attentive=> at-risk unresponsiveness"

■ No inconsistencies found.

Q23. Violation "Sleep or unresponsiveness => eyes closed & mouth open"

■ 3 inconsistencies found.

```
Running Q23. Violation "Sleep or unresponsiveness => eyes closed & mouth open"...
■ Inconsistencies found (1 rows)
(rdfLib.term.URIRef('http://test.org/in_cabin_domain#DriverState_62b16285b0174646b1b5cb64aeb5aec2'),
 rdfLib.term.URIRef('http://test.org/in_cabin_domain#sleep_instance'),
 rdfLib.term.URIRef('http://test.org/in_cabin_domain#undefined_atrisk_instance'),
 rdfLib.term.URIRef('http://test.org/in_cabin_domain#eye_state_2024_4_15_0_0_1'),
 rdfLib.term.URIRef('http://test.org/in_cabin_domain#mouth_state_2024_4_15_0_0_1'))
```

Q24. Is any state missing its threshold profile?

■ No inconsistencies found.

Q25. Are there any threshold profiles not used by an Actor State?

■ 7 inconsistencies found.

```
Running Q25. Are there any threshold profiles not used by an Actor State?...
■ Inconsistencies found (14 rows)
(rdfLib.term.URIRef('http://test.org/in_cabin_domain#tp_elderly_female_cold'),)
(rdfLib.term.URIRef('http://test.org/in_cabin_domain#tp_middle-aged_female_cold'),)
(rdfLib.term.URIRef('http://test.org/in_cabin_domain#tp_young_female_cold'),)
(rdfLib.term.URIRef('http://test.org/in_cabin_domain#tp_young_male_cold'),)
(rdfLib.term.URIRef('http://test.org/in_cabin_domain#tp_elderly_female_hot'),)
```

Q26. Detect future categories accidentally left unmapped

■ No inconsistencies found.

Q27. Show if new data falls outside modeled thresholds.

■ No inconsistencies found.

Q28. Check if new states appear without corresponding GCI or SWRL effects.

■ No inconsistencies found.

Q29. Future adaptability often breaks.

■ 7 inconsistencies found.

Running Q29. Future adaptability often breaks....

■ Inconsistencies found (18 rows)

```
(rdflib.term.URIRef('http://test.org/in_cabin_domain#tp_elderly_female_cold'),)
(rdflib.term.URIRef('http://test.org/in_cabin_domain#tp_elderly_male_cold'),)
(rdflib.term.URIRef('http://test.org/in_cabin_domain#tp_middle-aged_female_cold'),)
(rdflib.term.URIRef('http://test.org/in_cabin_domain#tp_middle-aged_male_cold'),)
(rdflib.term.URIRef('http://test.org/in_cabin_domain#tp_young_female_cold'),)
```

Q30. Unexpected extensions

■ 7 inconsistencies found.

Running Q30. Unexpected extensions...

■ Inconsistencies found (2246 rows)

```
(rdflib.term.URIRef('http://www.w3.org/2002/07/owl#Ontology'),)
(rdflib.term.URIRef('http://www.w3.org/2002/07/owl#ObjectProperty'),)
(rdflib.term.URIRef('http://www.w3.org/2002/07/owl#ObjectProperty'),)
(rdflib.term.URIRef('http://www.w3.org/2002/07/owl#ObjectProperty'),)
(rdflib.term.URIRef('http://www.w3.org/2002/07/owl#ObjectProperty'),)
```